OPEN RESEARCH OVERVIEW / VERSION 1.0.0

Intelligence Spillover, Generalization, Fluidity, Transfer, and Relay

ISGFTR | Theory-first research framework | Sizhe Tan

Core research idea

Intelligence may extend beyond the task, representation, or unit in which it was initially observed. ISGFTR examines five connected characteristics—spillover, generalization, fluidity, transfer, and relay—as candidate mechanisms for discovering, moving, testing, and retaining useful capabilities across contexts and systems. These are research hypotheses, not demonstrated AGI capabilities.

Conceptual architecture

Capability boundaries	Generative · Observed · Reachable · Evolvable
Discovery and adaptation	Structural folding · Directional intuition · Weak-evidence prospecting
Controlled intelligence flow	Selective permeability · Probe/validate gates · Counterevidence
System-level learning	Learning–execution decoupling · Spatial/temporal relay · Collective learning

Research collection and navigation

10 core papers — ISGFTR-001–010: boundaries, flow, transfer, system learning, relay, AGI limits, and open architecture

4 specifications — SPEC-001–004: capability/boundary, opportunity/evidence, validation lifecycle, and relay contracts

3 experimental cases — CASE-001–003: cross-domain LLM transfer, frozen-unit collective learning, weak-evidence prospecting

3 extensions — EXT-001–003: biomimetic fluidity, learned flow boundaries, evolvable intelligence potential

Validation and scientific boundaries

The roadmap calls for controlled baselines, explicit provenance, negative-transfer tests, independent replication, and clear separation of exploration permission from validated capability claims. Mobility does not establish utility; utility does not establish durable growth; recursive relay does not by itself establish recursive self-improvement or AGI.

Repository and research access

https://github.com/sizhet/Intelligence-Spillover-Generalization-Fluidity-Transfer-and-Relay

Release: v1.0.0 (theory-first) | License: Apache-2.0 | DOI: pending assignment | Invitation: critique, test, reproduce, and relay.